

Problem 1: Entropy

Background

Shannon entropy is defined as

$$H(X) = - \sum_{x \in X} p(x) \log p(x) \quad (\text{with the convention } 0 \log 0 := 0).$$

Problem 1A (Discrete case)

On each repeated trial of the same stimulus condition, a neuron produces a discrete response

$$X \in \{S, P, B\}$$

where S = silent, P = spike, B = burst. Let $p(X) = (p_S, p_P, p_B)$ be the PMF, with $p_S, p_P, p_B \geq 0$ and $p_S + p_P + p_B = 1$.

Now suppose recording tells you that the neuron is silent with probability

$$p_S = s \quad \text{for some fixed } 0 < s < 1,$$

but you are free to choose how the remaining probability $1 - s$ is split between P and B . Among all PMFs satisfying $p_S = s$, find:

1. the PMF(s) that *minimize* $H(X)$ and the minimum value,
2. the PMF that *maximizes* $H(X)$ and the maximum value.